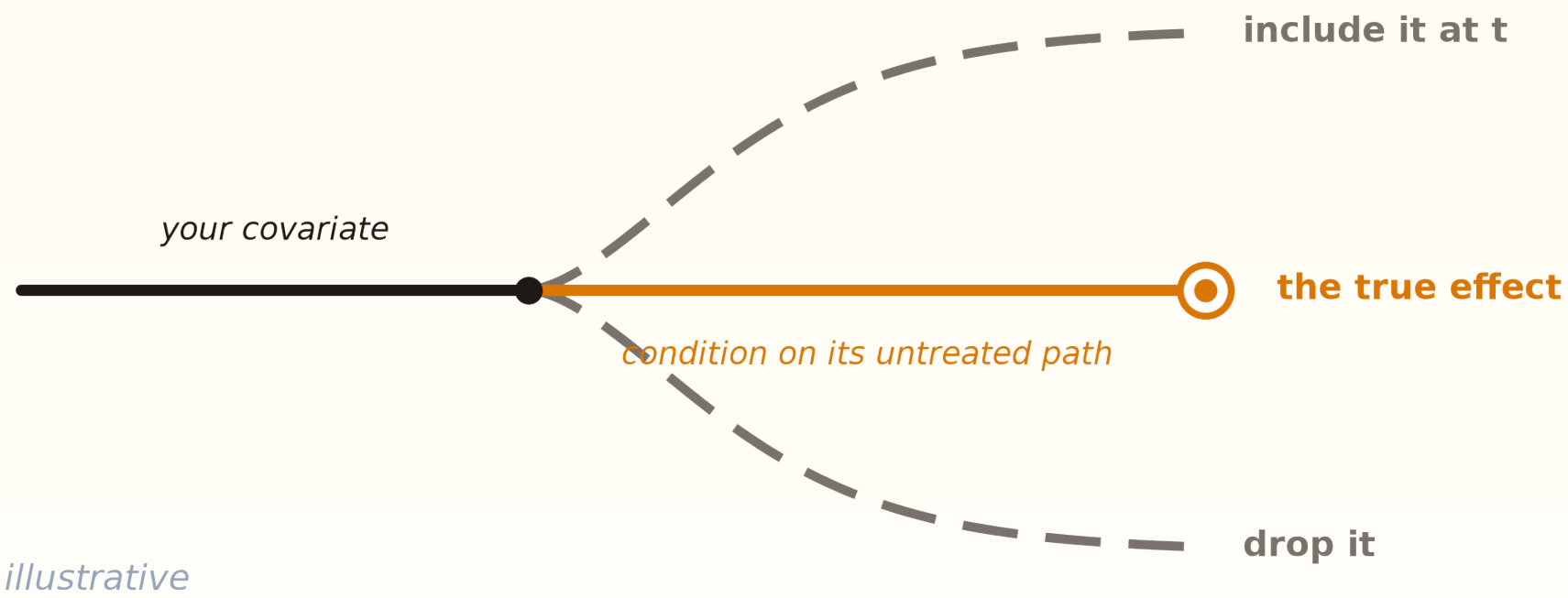
diff - diff

Include the covariate, or drop it?

Both can be wrong.



New in diff-diff 3.12: DiD with bad controls.

Caetano, Callaway, Payne & Sant'Anna (2026).

THE DILEMMA

A covariate that treatment can change.

Occupation. Hours worked. Health insurance. Firm size. Anything measured after treatment starts that treatment plausibly moves.

Caetano, Callaway, Payne & Sant'Anna (2026) call it a bad control when it matters for the untreated outcome trend (Condition 1) and treatment affects it (Condition 2).

Condition on it at t and you compare units at different values of its untreated version. Drop it and you throw away what its path says about the trend.

THE RECEIPT

Include it at t.

0.4886

reported SE 0.0142

TWFE with the bad control at t - unit-clustered

1.00

the true effect

simulated staggered panel - truth known by construction

Half the effect, netted out.

A simulated staggered panel built from the paper's first Monte Carlo design: 2,000 units, 4 periods, cohorts treated at t = 3 and 4, 974 never treated. Treatment moves the covariate by 0.50, and that half of the effect is exactly what the regression removes - the paper's own simulation reports the same bias (-0.501).

IT GETS WORSE

Drop it? Still wrong.

The covariate's untreated path carries information about the outcome trend. Discard the covariate and that information leaves with it.

In the paper's simulations, dropping the covariate is inconsistent in every design.

THE PAPER

Caetano, Callaway, Payne & Sant'Anna (2026): two ways out.

1 Condition on the pre-treatment value

under simple covariate unconfoundedness, the target reduces to Callaway-Sant'Anna with the covariate read at the base period - existing tools apply

2 Model the covariate's untreated path

covariate unconfoundedness given its pre-treatment value, observed confounders W and Z. W explains the covariate's path only - never parallel trends - and a doubly-robust, ML-ready score does the rest

Difference-in-differences with 'bad controls' - arXiv:2608.03881, an unpublished preprint (v2, September 2026).

ROUTE 1

Already in the box.

Read it at the base period.

On panel data, CallawaySantAnna reads every covariate at the cell's base period, never at t. Put the bad control in covariates and you have the paper's Approach 1 - no new code.

1.0540

SE 0.0235

covariates=["x", "z"] - W left out

misses the true 1.00 by two SE

0.9754

SE 0.0274

covariates=["x", "z", "w"] - W in

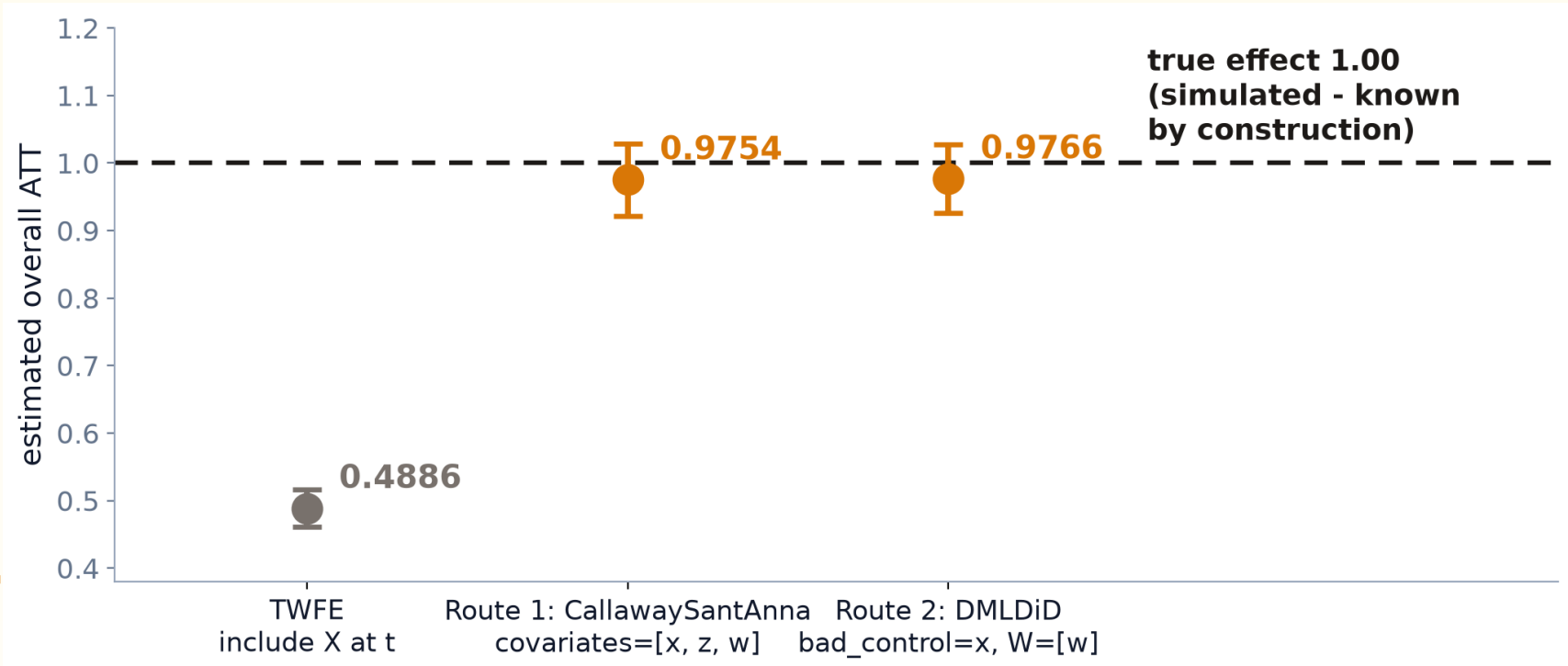
within one SE of the true 1.00

This design has an observed confounder W of the covariate's path. Leave it out and Route 1 misses; put it in and Route 1 works - but which conditioning set is right is an identification call, not a fit statistic. Simulated example, truth known by construction.

ROUTE 2

The bad-control lane.

What Route 2 buys you.



Both routes are valid. Route 1 puts W into parallel trends. Route 2 uses W only to explain the covariate's untreated path - so the lagged outcome can be W without becoming a parallel-trends variable (the paper's Remark 2) - plus a doubly-robust, ML-ready score and the pre-test on the next slide. Here: 0.9766 (SE 0.0260).

Seed-locked simulated example; whiskers are 1.96 analytical SE. On this draw both routes land within one SE of the truth - Route 2 is a different set of assumptions, not a bigger number.

THE PRE-TEST

Did treatment move the covariate?

Every cell also reports ATT_X(g,t): the effect of treatment on the bad control itself, with its own analytical SE - results.bad_control_summary().

cohort g	period t	ATT_X(g,t)	SE
3	2	**0.0019**	0.0166
3	3	**0.5137**	0.0156
3	4	**0.5055**	0.0299
4	2	**0.0120**	0.0169
4	3	**0.0061**	0.0190
4	4	**0.4764**	0.0263

Pre rows (t < g)

pre-test the path assumptions. They should be zero - and are, within SE.

Post rows (t >= g)

check that treatment moves the covariate - near the true 0.50.

A nonzero pre row flags a possible assumption violation - it is not evidence of a bad control. A zero post row does not prove treatment left the covariate alone. Simulated; truth known by construction.

THE CODE

Name it. Name its confounders.

```
res = DMLDiD(control_group="not_yet_treated", seed=0).fit(
    df, outcome="y", unit="unit", time="time",
    first_treat="first_treat",
    covariates=["z"],                  # Z: clean covariates
    bad_control="x",                   # the one treatment moves
    bad_control_covariates=["w"],      # W: confounders of its path
)
res.bad_control_summary()     # ATT_X(g,t) per cell
res.aggregate("event_study")

# Route 1 instead - read it at the base period:
CallawaySantAnna(control_group="not_yet_treated").fit(
    df, ..., covariates=["x", "z", "w"])
```

Pass the bad control through bad_control=, never also through covariates= (the fit rejects a column in both). seed= pins the folds.

IN THE WILD

Job displacement, NLSY79.

The paper's application: 748 displaced workers, 2,483 not. The bad control is the occupation score - displacement moves workers into occupations paying about 2.6% less (ATT_X -0.0261, SE 0.0092). The authors' estimates, not reproduced here:

almost -10%

naive TWFE on log earnings

with or without the occupation score

about -7%

ATT -0.0672, SE 0.0242

the paper's bad-control estimator on log earnings

Naive TWFE reads roughly 40% larger in magnitude. The bad-control estimator holds parallel trends conditional on the occupation a worker would have had without displacement - the covariate's untreated path.

Validated: numpy oracle of the paper's algorithm at 1e-12; tolerance goldens vs the authors' R badcontrols package, run black-box.

Lane scope: panel data only; cluster= yes, survey_design= no; no anticipation; varying base period. The headline ATT keeps the Callaway-Sant'Anna simple weighting, not the paper's Remark 4 overall.

diff - diff

Your covariate moved.

Your DiD can handle it now.

```
pip install diff-diff
```

Tutorial 33 reproduces every simulated number on this deck - both routes, the pre-test table, the event study and a ridge refit:

diff-diff.readthedocs.io/en/latest/tutorials/33_bad_controls.html

github.com/igerber/diff-diff

Caetano, Callaway, Payne & Sant'Anna (2026), arXiv:2608.03881 (preprint).